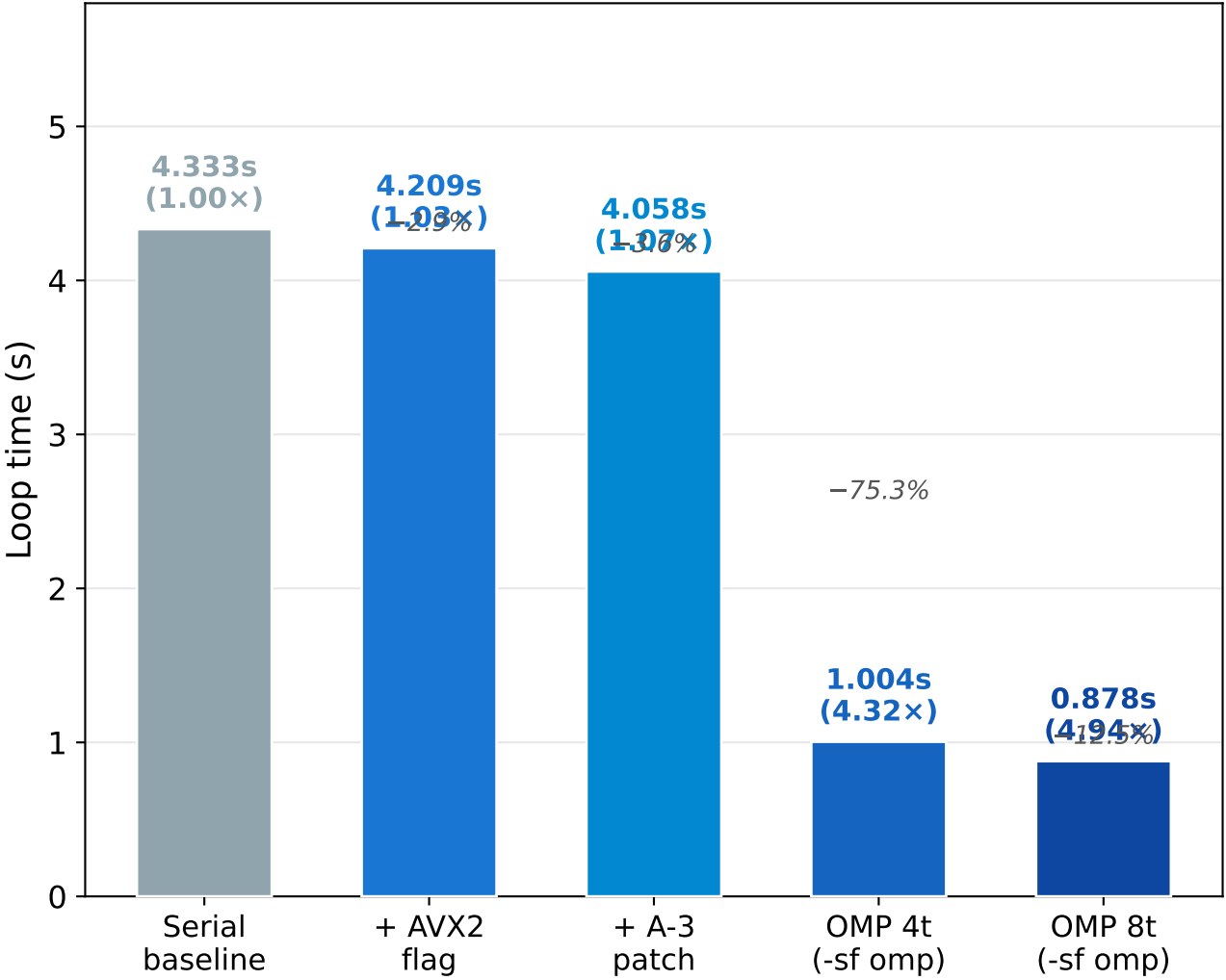
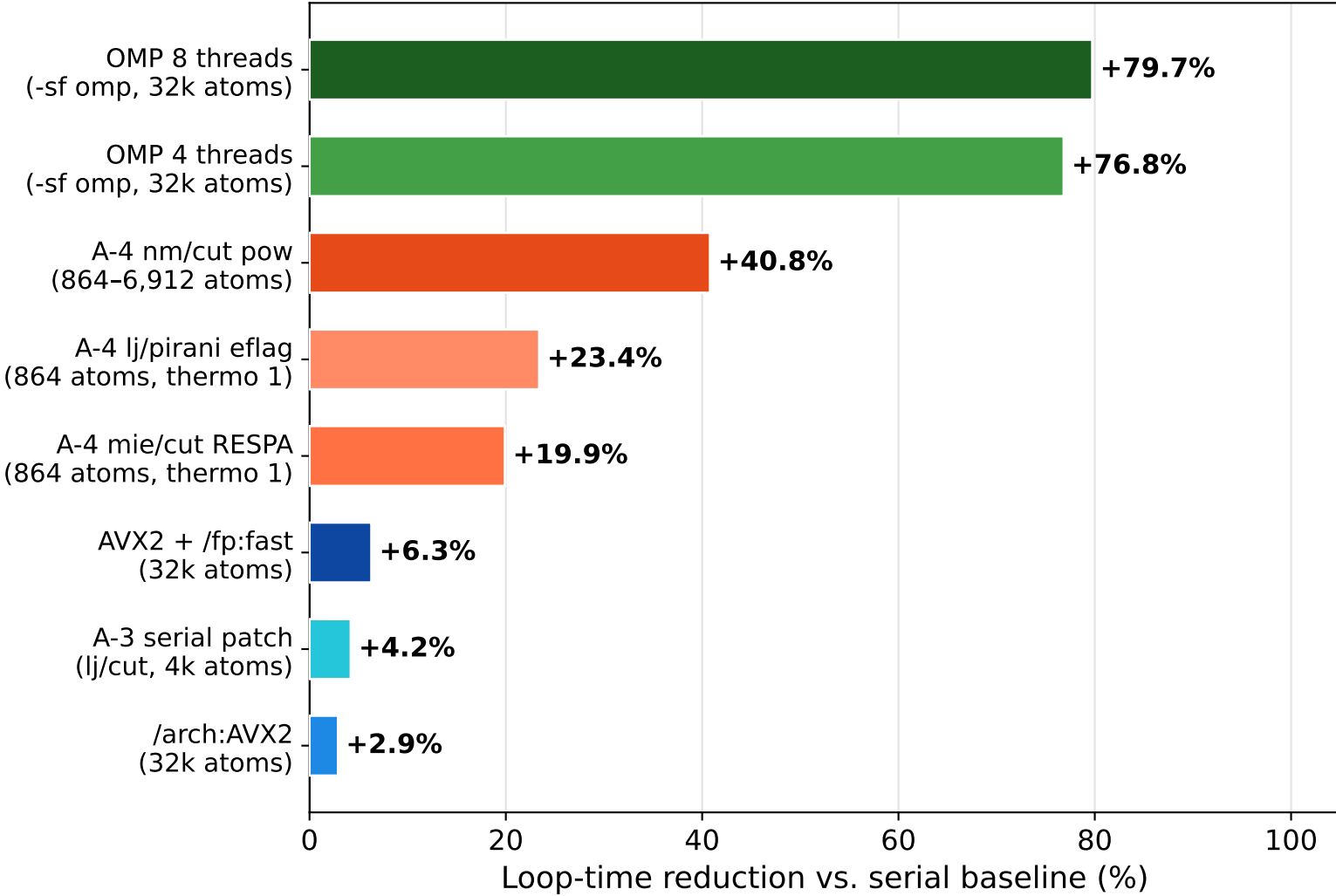


Figure 5 · Combined optimization summary

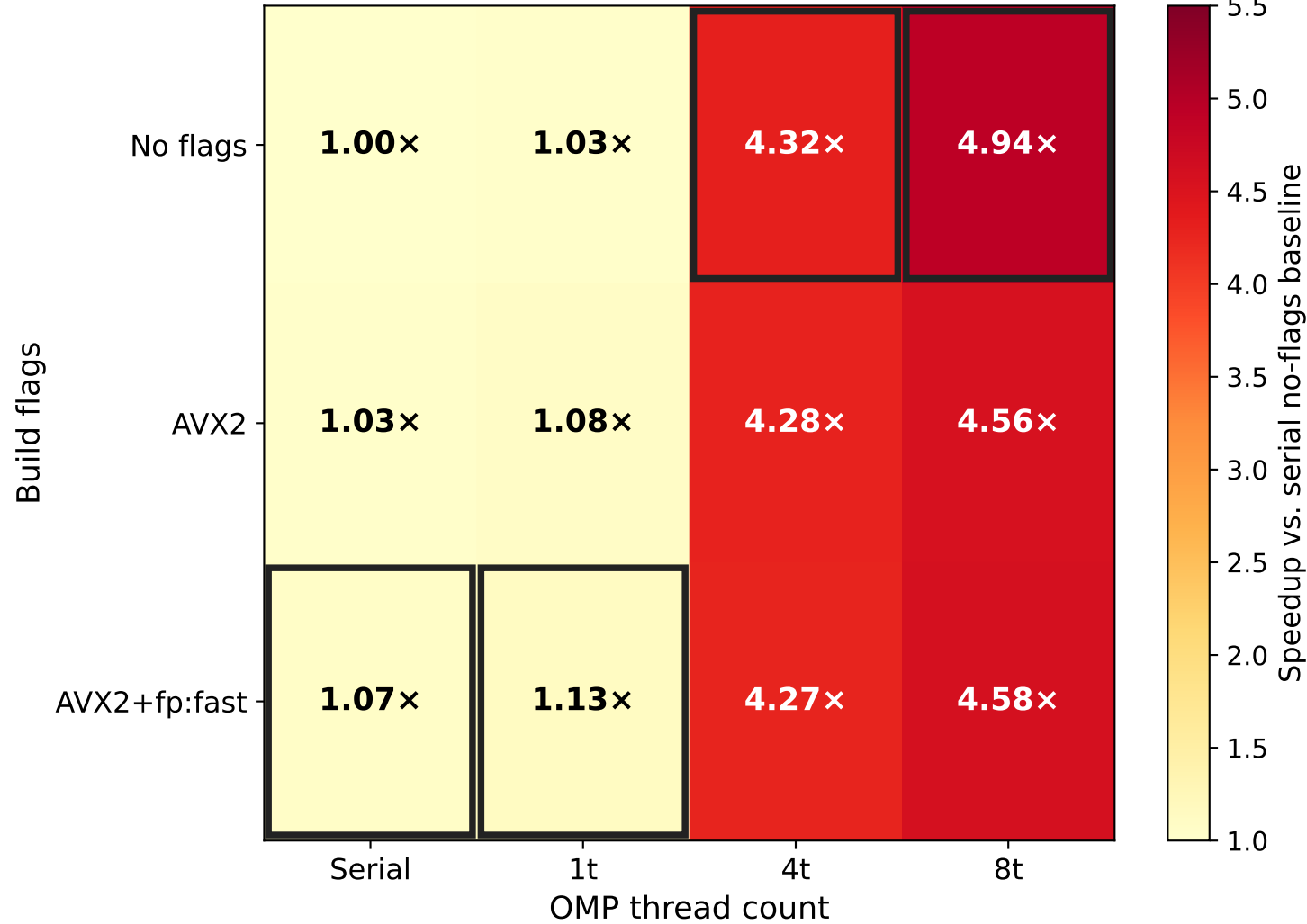
(a) Cumulative optimization waterfall
(32,000 atoms, 500 steps)



(b) Individual optimization contributions
(% loop-time reduction)



(c) Speedup heatmap — flags × threads
(■ = Pareto-optimal config)



(d) pair_style OMP coverage distribution
(51.1% → 62.5%, +11.4 pp)

